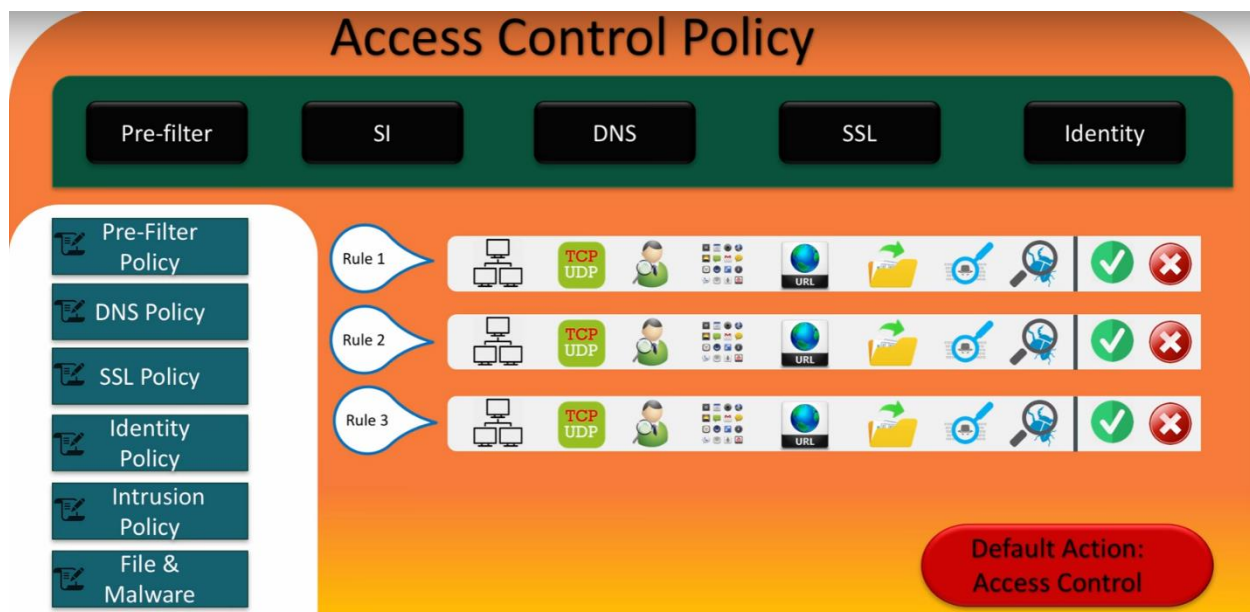


Access Control Policies:

- o Access Control Policies, or ACP's, are the Firepower rules that allow, deny & log traffic.
- o Access Policies are where you define your Firewall rule base and associated actions.
- o It is also the component that glues together most of the other policies we will look at.
- o The Firewall rules add in Firepower are more advanced than the old port-based rules.
- o Can include applications, user identity, zones, URLs and many other things in your rules.
- o In some ways, Access Control Policy rules are like traditional or Cisco ASA firewall rules.
- o They can match the traffic based on source or destination IP, as well as port number.
- o ACP's evaluate info, includes application, user, URL, traffic payload, business relevance.
- o The Access Control Policies (ACP) evaluate information this include, risk, and reputation.
- o ACP (Access Control Policies) tie many other policies together as well such as SSL policy.
- o Prefilter, SSL, Identity policies, Intrusion policies, and File policies are all used by the ACP.
- o ACP allows a hierarchy, with a base policy at the bottom, and child policies on the top.
- o Access Control Policy is a policy that pulls together rule sets for L3/L4,L7(Application).
- o ACP pulls together rule URL filtering, IPS/IDS, File/AMP (Advanced Malware Protection).
- o When evaluating conditions WITHIN a tab, multiple selections are treated as an OR.
- o When evaluating conditions BETWEEN tabs in Firepower it is considered as an AND.
- o Policies match IP prefixes, IP ports, the protocols TCP, UDP, and ICMP, and applications.
- o Nonmatching flows are dropped by default. Matching applications can only be denied.



ACP's can be found at **Policies -> Access Control -> Access Control**. Click New Policy and enter a **Name** and **Description**. If you have a base policy that you want to use as a parent, select it here. The Default Action applies when traffic does not match any rules. The action depends on what you're trying to achieve in your network. For example, an edge firewall will likely need to Block All Traffic. An internal device between networks may use Intrusion Protection as the default.

New Policy

Name:

Description:

Select Base Policy:

Default Action: ☒ Block all traffic ☐ Intrusion Prevention ☐ Network Discovery

Targeted Devices

Select devices to which you want to apply this policy.

Available Devices

☐ FTD-1

Selected Devices

☒ FTD-1

Prefilter Policy:

- o An Access Control List (ACL) check that runs before the Access Control Policy evaluation.
- o This allows or denies traffic without deep packet inspection, may improve performance.
- o ACL/Prefilter basic dictionary, block=deny, Fastpath=permit and Analyze=permit go to next.
- o Prefiltering is applied before Access Control in traffic flow, it operates at layer 4 like ACLs.
- o Prefiltering allows less control on how want to identify traffic and the actions to perform.

SSL Policy:

- o SSL Policy tells the Access Control Policy (ACP) how to handle the encrypted traffic.
- o This decrypt traffic for inspection, block encrypted traffic, or allow encrypted traffic.
- o Cisco Firepower has the ability to decrypt SSL traffic (or to be more accurate – TLS).
- o Depending on platform you are running this could significantly impact performance.

Identity Policy:

- o The Identify Policy used along with Realms to associate traffic with the users & Groups.
- o Cisco Firepower has the ability to integrate with user identities (Users/Groups AD etc).
- o Useful if want usernames in logs or if want to tie certain rules to specific users/groups.
- o In general, there are two ways of obtaining user identity in Firepower Passively, Actively.
- o Passively in Cisco Firepower learns the IP to user mappings from an external source.
- o This is typically either via a special agent which is running on a domain server AD Server.
- o Actively in Firepower can actively provide a login page to users to enter their credentials.

Intrusion Policies:

- o Intrusion Policy is where setup and configure your Intrusion Prevention/Detection policies.
- o It is based on Snort & like other similar systems it is signature-based protection mechanism.
- o Out of the box the Cisco FTD Firepower comes with a number of pre-defined base policies.

Malware and File Policies:

- o In Cisco firepower FTD the Malware and File policies allow you to perform two main tasks.
- o Firstly, through Malware and File polices can control the flow of files through the network.
- o secondly can check these for malware using Cisco's Advanced Malware Protection feature.
- o In Cisco Firepower FTD the File control can be based on the file type, protocol and direction.
- o wanted to prevent users downloading exe files, then this can be achieved with a file policy.
- o Advanced Malware Protection checks files for malware before passing files through firewall.

DNS Policy:

- o DNS-based Security Intelligence allows to whitelist, or blacklist traffic based on domain.
- o Cisco provides the domain name (FQDN) intelligence you can use to filter your traffic.
- o Can also configure custom lists and feeds of domain names tailored to your deployment.
- o Traffic blacklisted by DNS policy is blocked and not subject to any further inspection.
- o Also, not for intrusions, exploits, malware, and so on, but also not for network discovery.
- o Firewall sits between internal & external networks; it is well placed to inspect DNS traffic.
- o Basically, DNS Polices will drop any requests for known malicious domains included list.
- o DNS policies allow to statically whitelist and blacklist FQDNs as well as dynamically Cisco.

Mandatory and Default Rule:

There are two sections for rules; **Mandatory** and **Default**. These sections affect the evaluation of child policies. Child policies go between the two sections. Mandatory rules are evaluated first, then the child policy rules. Finally, the default rules.

Access Policy Action:

There are seven different actions which a rule can use in Access Policy in Cisco FTD.

Action	Description
Allow	Allows the traffic, there may yet be more inspections, such as Intrusion and File policies.
Trust	Sends traffic straight to the egress interface, without any extra inspections. Identity policies and rate limiting still apply.
Monitor	Logs traffic and continues to the rest of the rules.
Block	Drops traffic silently, causing the connection to timeout.
Block with reset	Drops traffic and sends a TCP FIN message to client so the connection closes rather than times out.
Interactive Block	Displays a web page with conditions that users may accept. This is where the Interactive Block Response Page comes into play.
Interactive Block with Reset	Combination of interactive block, with a TCP FIN message to client so the connection is close rather than times out.

Zones	The Zones tab allows Firepower to match traffic based on the source or destination zone. Zones are objects that contain one or more interfaces.
Networks	The Networks tab uses IP addresses, much like a common ACL.
Geolocation	This can match traffic from a particular country or region.
VLAN	The VLAN tab matches traffic based on VLAN.
Applications	The Applications tab allows traffic matching based on the application.
Ports	It's where you can match traffic based on source and destination ports.
URL	On the URL tab, we can match traffic to URL. This may use a list of specific URL's, or category of URL's.
SGT/ISE Attributes	The SGT/ISE Attributes tab can match traffic based on the users Security Group Tag (SGT).
Inspection	The Inspection tab is where to assign the Intrusion and File policies. If traffic matches rule & action is allow, then firepower evaluates these.
Logging	Each connection can log data to the event viewer. This is how we get information into the Dashboard and Analysis sections.